

MATHEMATICAL THERMAL MODEL PERFORMANCE REPORT

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PURPOSE

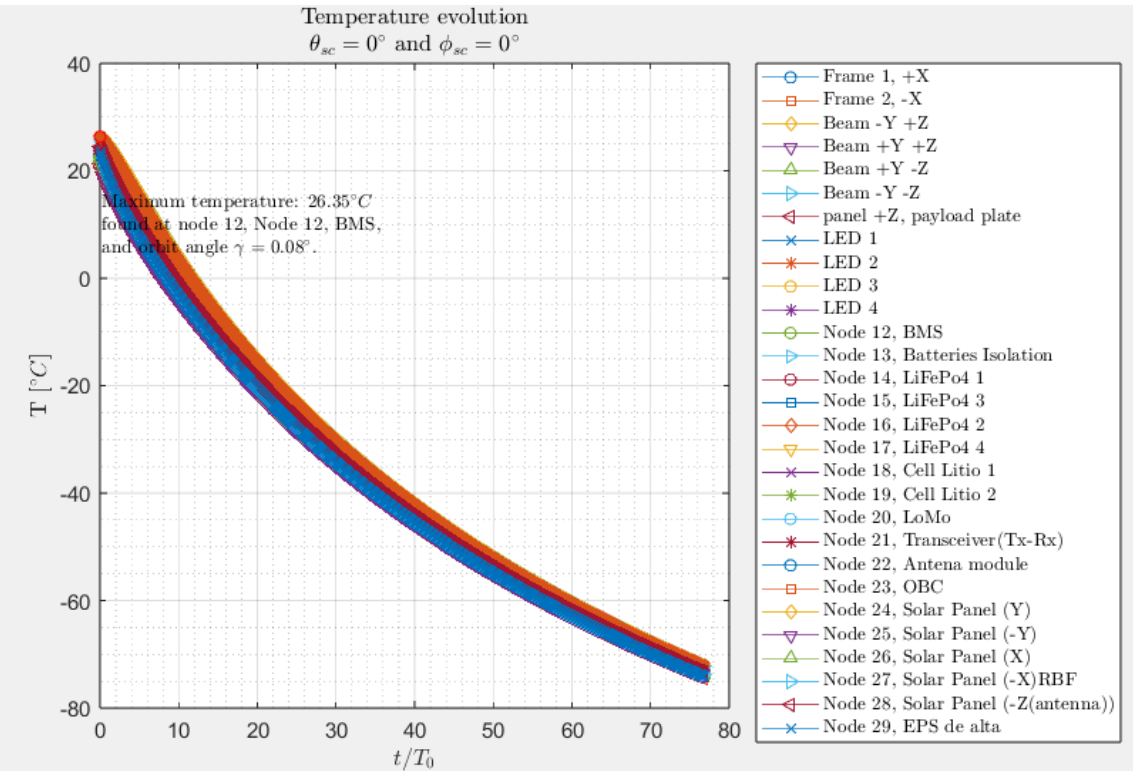
This document aims to compile the tests performed on the mathematical thermal model to test its performance as well as their respective conclusions.

QUALITATIVE TESTS

Tests performed to demonstrate the general behaviour of the model without taking into consideration the exact value of the results. Only the general tendencies are observed in these tests.

COOLING

The satellite was simulated with all systems off in a completely void and cold environment. An initial temperature of 300K was set.

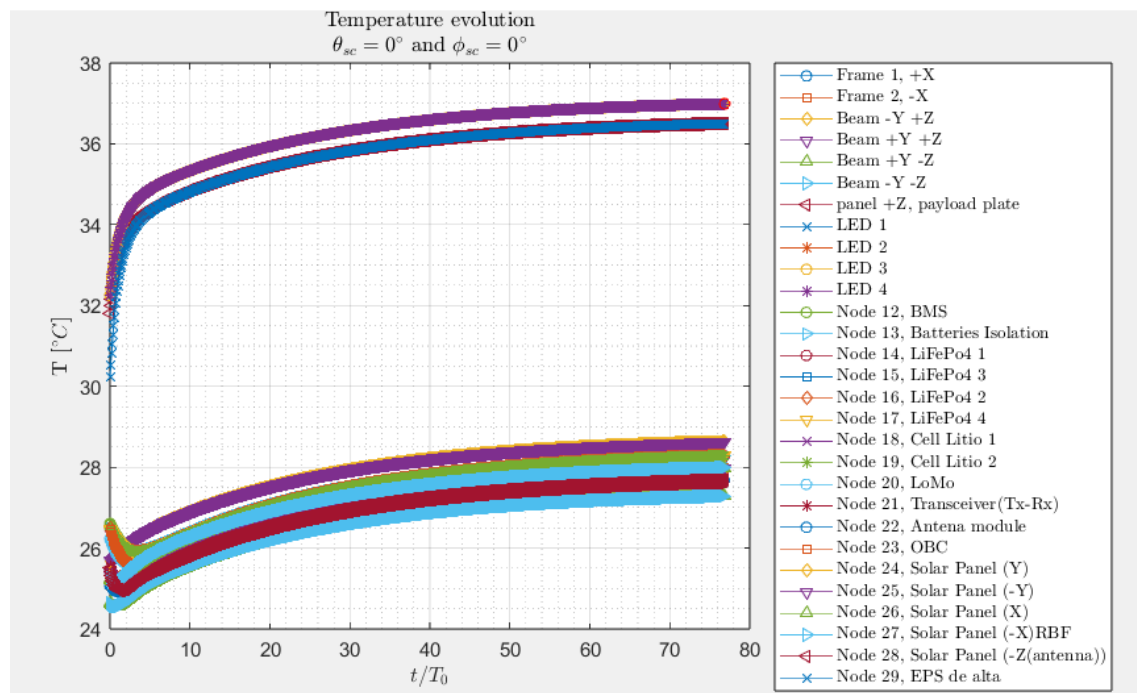


As expected, the temperature dropped due to radiation emission, at a rate that slowed down as the satellite cooled.

This is consistent with theory.

HEATING AND STATIONARY REGIME

The satellite's LEDs were turned on permanently in a cold void environment. The total heat generated was 12 W.



STATIONARY TENDENCY

The temperatures fluctuate before asymptotically approaching some fixed values. This is expected of a transition into stationary regime, approaching the values at which the generated heat is the same as the radiated heat.

This is consistent with theory

INTERNAL CONDUCTION

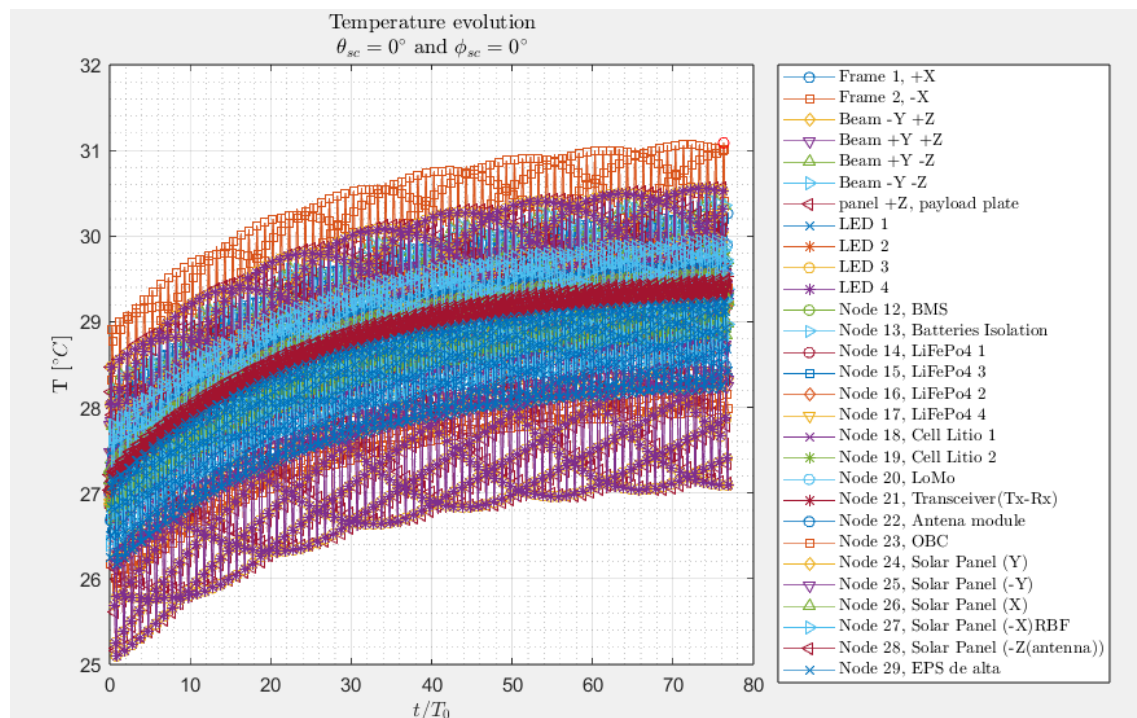
Inspecting the temperatures closer with the plot browser tool, we can identify that reached stationary regime the temperatures align such that they increase as the components get closer to the heat generating elements (The LEDs) with the LEDs being at the highest temperature. This is consistent with theory as the heat flows from higher to lower temperature.

We can see that the nodes separate into two groups, with the nodes integrated into the payload plate at considerably higher temperatures than the rest of the satellite. This indicates that the conductance between the payload plate and the rest of the satellite is considerably low.

Regardless this behaviour is consistent with theory.

SOLAR CYCLE AND ALBEDO

With the satellite off, we simulate the sun radiation and albedo from earth.

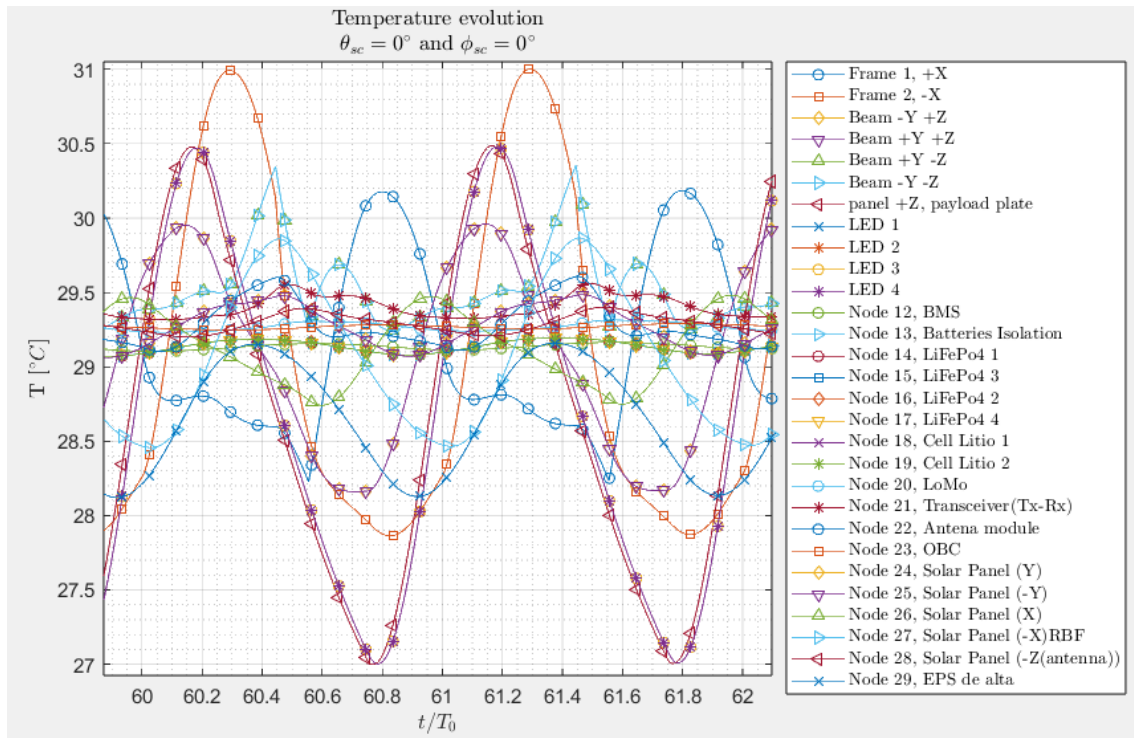


EQUILIBRIUM TENDENCY

We can observe how the satellite's overall temperature increases asymptotically approaching equilibrium with the environment, when the energy radiated per solar cycle equals the energy absorbed.

This is consistent with theory.

TEMPERATURE FLUCTUATIONS



A closer inspection with the plot browser tool reveals that those nodes closer to the surface (Solar panels, rails, beams...) fluctuate with higher amplitude than those closer to the centre (Batteries, computers...). This is expected as the surface nodes are more exposed to the sun, as well as being the ones that radiate the heat.

This is consistent with theory.